

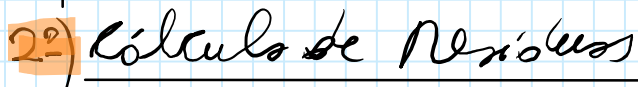
domingo, 7 de septiembre de 2025 00:17

6) Sintetice un cuadripolo que cumpla con:

$$Y_{21} = \frac{s(s^2 + 1)}{(s^2 + 2)(s^2 + 5)}$$

Produzenten (als die TCS), Recordon $\left\{ \begin{array}{l} y_{22} = \frac{z_{11}}{\text{Det}(C)} \\ y_{21} = \frac{z_{21}}{\text{Det}(C)} \end{array} \right.$

4º) Métodos físicos de síntesis y remoción



$$\bullet Z_4(b) = \frac{(b^2+2)(b^2+5)}{34(b^2+7/3)} - \frac{1}{b} = \frac{(b^2+2)(b^2+5) - 3(b^2+7/3)}{34(b^2+7/3)} = \frac{b^4+7b^2+10-3b^2-7}{34(b^2+7/3)} = \frac{b^4+4b^2+3}{34(b^2+7/3)}$$

$$Z_4(b) = \frac{(b^2+1)(b^2+3)}{38(b^2+7/3)}, \quad Y_4(b) = \frac{38(b^2+7/3)}{(b^2+1)(b^2+3)}$$

$$3f(f^2+7f+3)$$

$$(f^2+1)(f^2+3)$$

Ejercicio 6 - Hoja 2

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$$Y_6(s) = Y_4(s) - \frac{2K_1 \cdot s}{(s^2 + 7(3))}, \quad 2K_1 = \lim_{s \rightarrow -1} Y_4(s) \cdot \frac{(s^2 + 7(3))}{s} = \lim_{s \rightarrow -1} \frac{3s(s^2 + 7(3))}{(s^2 + 1)(s^2 + 3)} \cdot \frac{(s^2 + 7(3))}{s}$$

$$2K_1 = \lim_{s \rightarrow -1} \frac{3(s^2 + 7(3))}{(s^2 + 1)(s^2 + 3)} = \frac{3(-1 + 7(3))}{-1 + 3} = 2 = 2K_1$$

$$Y_6(s) = \frac{3s(s^2 + 7(3))}{(s^2 + 1)(s^2 + 3)} - \frac{2 \cdot s}{(s^2 + 1)} = \frac{3s(s^2 + 7(3)) - 2s(s^2 + 3)}{(s^2 + 1)(s^2 + 3)} = \frac{3s^3 + 7s - 2s^3 - 6s}{(s^2 + 1)(s^2 + 3)}$$

$$Y_6(s) = \frac{s(s^2 + 1)}{(s^2 + 1)(s^2 + 3)} = \frac{s}{(s^2 + 3)} = Y_6(s), \quad \frac{s^2 + 3}{s} = 36(s)$$

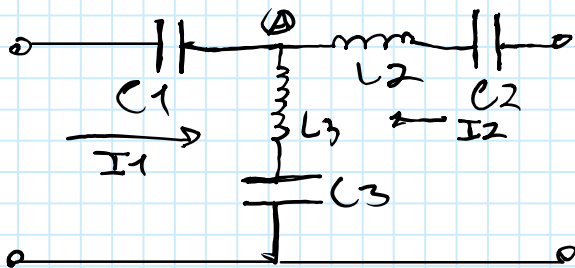
$$Z_8(s) = Z_6(s) - K_{\infty} \cdot s, \quad K_{\infty} = \lim_{s \rightarrow \infty} \frac{Z_6(s)}{s} = \lim_{s \rightarrow \infty} \frac{s^2 + 3}{s} \cdot \frac{1}{s} = 1 = K_{\infty}$$

$$Z_8(s) = \frac{s^2 + 3}{s} - 1 \cdot s = \frac{s^2 + 3 - s^2}{s} = \frac{3}{s} = Z_8(s)$$

$$Z_{10}(s) = Z_8(s) - \frac{K_0}{s}, \quad K_0 = \lim_{s \rightarrow 0} Z_8(s) \cdot s = \lim_{s \rightarrow 0} \frac{3}{s} \cdot s = 3 = K_0$$

$$Z_{10}(s) = \frac{3}{s} - \frac{3}{s} = 0 \Rightarrow \text{cable}$$

30) Circuito



$$C_1 = \frac{1}{K_0} = 1, \quad C_2 = \frac{1}{K_0} = 1/3, \quad L_2 = K_{\infty} = 1$$

$$Y_{L3, C3} = \frac{1}{\frac{1}{C_3 s} + s L_3} = \frac{1}{\frac{1}{s C_3} + s L_3} \cdot \frac{s C_3}{s C_3} = \frac{s C_3}{s^2 L_3 C_3 + 1}$$

$$Y_{L3, C3} = \frac{s C_3}{s^2 L_3 C_3 + 1} \cdot \frac{1/L_3 L_3}{1/L_3 L_3} = \frac{s (1/L_3)}{s^2 + \frac{1}{L_3 C_3}}$$

$$\frac{2K_1 \cdot s}{s^2 + 1} \Rightarrow \begin{cases} \frac{1}{L_3} = 2 \\ \frac{1}{L_3 C_3} = 1 \end{cases} \Rightarrow \begin{cases} L_3 = 1/2 \\ C_3 = 2 \end{cases} \Rightarrow \frac{1}{(1/2) \cdot 2} = 1 \Rightarrow 2 = C_3$$

40) Verificación

Recordamos que $-\frac{I_2}{I_1} \Big|_{V_2=0} = T(s) = \frac{-Y_{21}}{Y_{11}} \Rightarrow$ Buscamos el sistema con motores de $Y(s)$

$$\begin{bmatrix} 1 & Z_{C1} \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 \\ Y_{L3, C3} & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & Z_{L2, C2} \\ 0 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 + Z_{C1} Y_{L3, C3} & 1 \cdot 0 + Z_{C1} \\ 0 \cdot 1 + 1 \cdot Y_{L3, C3} & 0 \cdot 0 + 1 \cdot 1 \end{bmatrix} \begin{bmatrix} 1 & Z_{L2, C2} \\ 0 & 1 \end{bmatrix}$$

$$[T] \Rightarrow \begin{bmatrix} 1 + Z_{C1} Y_{L3, C3} + 0 & (1 + Z_{C1} Y_{L3, C3}) \cdot Z_{L2, C2} + Z_{C1} \\ Y_{L3, C3} \cdot 1 + 1 \cdot 0 & Y_{L3, C3} \cdot Z_{L2, C2} + 1 \cdot 1 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$$

$$D = Y_{L3, C3} \cdot Z_{L2, C2} + 1 = \frac{s (1/L_3)}{s^2 + \frac{1}{L_3 C_3}} \cdot \left(\frac{1}{C_2 s} + L_2 s \right) + 1 = \frac{s \cdot 2}{s^2 + 1} \cdot \left(\frac{3}{s} + s \right) + 1 = \frac{2s}{s^2 + 1} \cdot \frac{3 + s^2}{s} + 1 = \frac{2(3 + s^2) + (s^2 + 1)}{s^2 + 1} = \frac{6 + 2s^2 + s^2 + 1}{s^2 + 1} = \frac{3s^2 + 7}{s^2 + 1} = 3 \cdot \frac{(s^2 + 7(3))}{s^2 + 1} \Rightarrow \frac{1}{D} = \frac{(s^2 + 1)}{3(s^2 + 7(3))} = -\frac{Y_{21}}{Y_{11}} = T(s) \checkmark$$

$$D = \frac{2(3+s^2) + (s^2+1)}{s^2+1} = \frac{6+2s^2+s^2+1}{s^2+1} = \frac{3s^2+7}{s^2+1} = \frac{3(s^2+7/3)}{s^2+1} \Rightarrow \frac{1}{D} = \frac{(s^2+1)}{3(s^2+7/3)} = \frac{-1/21}{1/11} = 7/11 \checkmark$$